

Amit Shenoy

Boston, MA || 508-864-5532 || ashenoycompany@gmail.com

LinkedIn: itsamit || GitHub: xX-its-amit-Xx || *Mission: Solve as many puzzles as possible*

EDUCATION

Northeastern University

Fall 2026 – Expected May 2027

Incoming M.S., Bioengineering (Thesis-Track)

Advisor: Dr. Benjamin Gyori, INDRA-CoGEx Lab

Northeastern University

Sept 2021 – Dec 2025

B.S., Bioengineering (Computational, Systems, and Synthetic Biology track), Minor: Mathematics

GPA: 3.91

Honors: Honors Program; PEAK Research Grants; Dean’s List (all semesters); LEAD360 Emerging Leader

RESEARCH & PROFESSIONAL EXPERIENCE

COMBINE Lab, Northeastern University (PI: Dr. Mona Minkara), Boston, MA

May 2023 – Dec 2025

Undergraduate Researcher

- Built reproducible HPC pipelines (SLURM) for large-scale molecular modeling, running Schrödinger-suite docking (Glide, Prime) and MM-GBSA free-energy calculations across thousands of protein–ligand systems.
- Designed and automated job-orchestration and QC checkpoints, cutting manual pipeline intervention by >80% and standardizing lab-wide reproducibility practices.
- Contributed structural biophysics and computational analysis to two manuscripts currently in preparation.
- Delivered 15+ technical talks to lab, department, and conference audiences on pipeline design and molecular modeling results.
- Mentored newer lab members on SLURM job submission, Schrödinger workflows, and basic scripting practices.

UCB Biosciences, Cambridge, MA

Jan – Dec 2025

Computational Data Science Co-op

- Processed large multimodal datasets (chemical structures, assay readouts, sample metadata) supporting drug-discovery decision-making across a full-year embedded team placement.
- Developed statistical and machine learning models to prioritize compounds, narrowing the screening space to <1% high-confidence candidates.
- Applied UMAP and SHAP-based interpretability methods to flag data artifacts and mislabeled samples, improving model reliability and analyst trust in outputs.
- Collaborated with experimental and data science teams to translate model outputs into actionable compound-prioritization decisions.

Arbor Biotechnologies, Cambridge, MA

Jan – Jun 2024

Gene Therapy Co-op

- Designed Design-of-Experiments (DOE) studies to optimize AAV production in mammalian and suspension systems, improving system productivity by >250%.
- Applied statistical analysis to qPCR and ddPCR titer data to identify the process parameters driving performance gains.
- Worked cross-functionally with process development and analytical teams to translate findings into standard operating procedure changes.

SELECTED PROJECTS (GitHub: xX-its-amit-Xx)

OpenADMET-PXR

2026

- Solo entry (top 100) in the Hugging Face OpenADMET competition, predicting binding affinity (pEC50) for the Pregnane X Receptor across 513 held-out compounds trained on ~4,000 labeled examples.
- Iterated through 35+ modeling approaches: ChemBERTa fine-tuning, ECFP+LightGBM baselines, hybrid transformer architectures fusing protein and compound embeddings, and transfer learning across analogous assays. Current Pearson $r \approx 0.63$.
- Actively developing activity-cliff-aware pretraining: training a generalized model on structural-activity-cliff pairs to sensitize representations to small structural changes that drive large activity shifts, then fine-tuning on sparse PXR data.

OmicsQC & scRNA-Flow

2026

- Built end-to-end sequencing QC and scRNA-seq analysis pipelines (Scanpy/AnnData): filtering, normalization, doublet detection, batch integration, clustering, and marker-gene analysis.
- Implemented automated detection of low-quality cells, ambient RNA contamination, and batch effects, packaged with reproducible HTML/Markdown QC reports.

ATAC-QC

2026

- Developed an ATAC-seq QC and peak-analysis pipeline with TSS enrichment scoring, fragment-size distribution diagnostics, and nucleosome banding checks.

BayesBio

2025

- Built a Bayesian/MCMC pipeline to infer biological model parameters from noisy experimental data, returning calibrated uncertainty estimates rather than point values.

CONFERENCE PRESENTATIONS

- **AAAS Annual Meeting** (American Association for the Advancement of Science) – 2025, Presenter
- **RISE: Research, Innovation, and Scholarship Expo**, Northeastern University – 2024, 2025
- **MBN (Massachusetts Biophysical Network) Biophysics Conference** – 2024
- **NESS Poster Symposium** – Attendee
- **BSCP Conference** – Attendee
- **Source Fair**, Northeastern University – Presenter

30+ scientific presentations total across class, lab, and co-op settings.

MANUSCRIPTS IN PREPARATION

- Two manuscripts in preparation with the COMBINE Lab (Northeastern University), covering computational pipeline design and structural biophysics analysis.

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, pandas, scikit-learn, PyTorch, Jupyter), R, SQL (PostgreSQL), Bash, C++, MATLAB, Kotlin

Data Analysis & ML: Statistical modeling, Bayesian inference, dimensionality reduction (PCA, UMAP), SHAP, classification/regression, transformer fine-tuning, transfer learning

Bioinformatics: NGS data processing (FASTQ, BAM), sequencing QC, scRNA-seq, ATAC-seq, RNA-seq

Cheminformatics: RDKit, Morgan/ECFP fingerprints, QSAR modeling, ADMET prediction, ChemBERTa, activity-cliff analysis, applicability domain

Biophysics: Schrödinger suite (Maestro, Glide, Prime), induced-fit docking, MM-GBSA, BPMD, molecular dynamics, VMD, PyMOL

Wet Lab: AAV production (mammalian & suspension), primer design, qPCR, ddPCR, fluorescence microscopy, western blot, TapeStation, RNA/DNA extraction, CRISPR/dCas9, Gibson assembly, site-directed mutagenesis, Golden Gate assembly, colony PCR

Tools: Git, Docker, Conda, SLURM, Jupyter, Benchling, AutoCAD, SolidWorks, Arduino

AI-Native Workflow: Cursor, Claude Code, Windsurf, sub-agent orchestration, AGENTS.md, MCP servers

LEADERSHIP & SERVICE

- **Co-founder & Co-President**, GNU@NU – libre/open-source software advocacy community, Northeastern University
- **Co-Chair**, NU Sustainability Committee
- **AVP Sustainability**, Student Government Association (SGA), Northeastern University
- **Project Manager / Co-Founder**, Ambassadors for Change
- **Science Fair Judge**, Massachusetts State Science & Engineering Fair (MSEF) – school and regional level
- **Mentor**, Bridge to Calculus (Northeastern-sponsored math mentoring program)
- **Programming Mentor**, CodeWiz
- **STEM Tutor**, private middle school and self-employed (elementary–high school)
- Library volunteer

INTERESTS

Piano, Chess, Tennis, Game Development, Greek Mythology, Social Deduction Games (Blood on the Clocktower, One Night Ultimate Werewolf), *Death Note*